

Member ID: _____

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SQL DATABASE FUNDAMENTALS (345) **STATE 2024**

Multiple Choice:

50 @ 2 points each

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. The wildcard characters of the percent sign and the underscore (% , _) are used to specify pattern in a column for the operator _____.
 - A. WITH
 - B. HAVING
 - C. LIKE
 - D. AS
2. Which of the following is NOT an ANSI standard command?
 - A. UNION
 - B. ROLLBACK
 - C. HAVING
 - D. TRIM
3. A SQL function that allows the user to access a specific set of records from a database table is called a(n) SQL _____.
 - A. statement
 - B. access
 - C. clause
 - D. operator
4. Which of the following conditions will NOT check if field1 has a value of "OFF"?
 - A. SELECT * FROM table WHERE field1 == "OFF"
 - B. SELECT * FROM table WHERE NOT field1 != "OFF"
 - C. SELECT * FROM table WHERE field1 IN ("OFF")
 - D. SELECT * FROM table WHERE field1 >= "OFF" AND field1 <= "OFF"
5. A full outer join between table1 and table2 is equivalent to _____.
 - A. FULL OUTER join is equivalent to a CROSS join between the two tables
 - B. All Records from table1 and table2 that they don't share
 - C. Only data that matches in both tables
 - D. All data that matches in both tables, or exists in either table
6. Which statement would get the records where field1 has a value between 100 and 200 (not including 100 and 200)?
 - A. WHERE field1 > 100 && field1 <200
 - B. WHERE field1 BETWEEN 100 AND 200
 - C. WHERE field1 < 200 AND field1 > 100
 - D. WHERE field1 BEWEEN(101, 199)

7. A LIMIT clause is important because it improves efficiency by restricting how many records the database has to access.
 - A. TRUE
 - B. FALSE
8. The order of operations for logical operators in SQL is
 - A. AND then NOT then OR
 - B. NOT then AND then OR
 - C. NOT then AND / OR (same Level)
 - D. NOT / AND / OR (all same Level)
9. The query SELECT 1885 & 170 OR TRUE?
 - A. 8
 - B. 170
 - C. 1
 - D. 0
10. Which of the following is a valid data type?
 - A. DATE
 - B. STRING
 - C. BOOLEAN
 - D. DOUBLE
11. Why would this query fail - CREATE TABLE table1 (int personID, char yesNo)
 - A. No primary key is listed.
 - B. Char is not a valid data type.
 - C. Datatype and field name not in the correct order
 - D. Field names must be lowercase
12. Which Query will NOT find all rows where the character field, field1 is 'a'?
 - A. SELECT * FROM table1 WHERE field1 like 'a';
 - B. SELECT * FROM table1 WHERE field1 in (SELECT 'a')
 - C. SELECT * FROM table1 WHERE field1 < 'b'
 - D. SELECT * FROM table1 WHERE field1 = 65
13. Which is NOT an advantage of a normalized database?
 - A. Reduces data storage
 - B. Enforces data consistency
 - C. Reduces the average number of joins to access data
 - D. Encourages quicker data access

14. To list the number of tables in a database you would use the ____ command.
- A. LIST TABLES
 - B. SHOW TABLES
 - C. DISPLAY TABLES
 - D. PRESENT TABLES
15. What would be the outcome of the following SELECT statement - SELECT field1, "*", field2 FROM table;
- A. All the values in field1, followed by all the values from each field in the table, followed by the values in field2
 - B. All the values from each field in the table with first1 as the first column and field2 as the last column
 - C. All the values in each field in the table, excluding field1 and field 2
 - D. All the values in field1, *, all the values in field2
16. Database normalization rules were established in what decade?
- A. 1950s
 - B. 1960s
 - C. 1970s
 - D. 1980s
17. Which of the following statements is true about a UNION ALL statement?
- A. The resulting data will come out sorted
 - B. The resulting data will be free from duplicates
 - C. UNION ALL is faster than UNION
 - D. UNION ALL can take a list of tables to combine
18. The acronym that stands for the process of manipulating data using insert, update, and delete is ____
- A. DML
 - B. DDM
 - C. CMD
 - D. UML
19. A Primary Key consisting of multiple fields is known as a ____.
- A. Alternate Key
 - B. Super Key
 - C. Candidate Key
 - D. Composite Key

20. How many normalization forms are there?
- A. 4
 - B. 5
 - C. 6
 - D. 7
21. Which normalization form states that an entity should depend solely on the unique identifier of the entity?
- A. 1NF
 - B. 2NF
 - C. 3NF
 - D. 4NF
22. When running a query, the aggregate function COUNT(field1) includes entries where field1 is NULL.
- A. TRUE
 - B. FALSE
23. The transaction property stating a transaction will either be performed in its entirety or not at all is known as ____.
- A. Reliability
 - B. Atomicity
 - C. Durability
 - D. Consistency
24. The acronym for the standard properties of a transaction is ____.
- A. DARC
 - B. ACID
 - C. CARD
 - D. FLAC
25. An intersection is the result of what type of join?
- A. CROSS
 - B. OUTER
 - C. INNER
 - D. FULL
26. Which WHERE clause would check if the field1 contains "apple", "pear", or "grape"?
- A. WHERE field1 IN ("apple", "pear", "grape ")
 - B. WHERE field1 = "apple", "pear", "grape"
 - C. WHERE field1 CONTAINS ("apple", "pear", "grape")
 - D. WHERE field1 = "apple" AND field1 = "pear" AND field1 = "grape"

27. An SQL Trigger is used to ____.
- Enforce business rules
 - Speed up the time it takes to run a query
 - Manage who has access to the database
 - Reduce complexity during development
28. An Autoincrement field in SQL ____.
- Assigns random numbers to a key field of a table
 - Is required if a table does not have a primary key
 - Puts a table in 2nd normal form
 - Is often a surrogate key

Consider the following three tables for questions 29-37

Movie				MovieActor		Actor		
movieId	title	genreId	genre	movieId	actorId	actorId	name	dob
1	Speed	1	Horror	1	1	1	Bob	1950-01-01
2	Strength	2	War	1	3	2	Sally	1970-05-11
4	Size	4	Drama	1	6	3	Jane	2000-07-12
8	Shape	2	War	2	4	4	Carl	1970-05-11
				2	2	5	Jim	1999-06-01
				8	6	6	Sally	1994-06-24
				8	4			

29. How many INSERT queries would be necessary to save “Tony” was in the movie “Tiny”
- 1
 - 2
 - 3
 - 4
30. What level of normalization is table Movie?
- It does not pass any normalization forms
 - 1NF
 - 2NF
 - 3NF

31. What would be the order of the actorId given the following query:
SELECT DISTINCT c.actorID FROM Movie a
INNER JOIN MovieActor b ON a.movieId = b.movieId
INNER JOIN Actor c ON b.actorId = c. actorId
WHERE genreId = 1 ORDER by name DESC, dob ASC
- A. 6,3,1
 - B. 1,3,6
 - C. 3,6,1
 - D. 1,6,3
32. Which JOIN & WHERE clause would produce Actors who were not in any movies?
SELECT name FROM actor a
- A. LEFT JOIN MovieActor b ON a.actorId = b.actorId WHERE b.movieId IS NULL
 - B. INNER JOIN MovieActor b ON a.actorID = b.actorId WHERE b.movieID IS NULL
 - C. RIGHT JOIN MovieActor b ON a.actorID = b.actorId WHERE b.movieID IS NULL
 - D. Can't join if there are no actors to join to.
33. Which query would produce the youngest actor / actress and the movie they were in?
SELECT title, name, dob FROM Movie
- A. INNER JOIN Actor b ON a.movieId = b.movieId WHERE dob = MAX(dob)
 - B. INNER JOIN MovieActor b ON a.movieId = b.movieId INNER JOIN Actor c ON b.actorId = c.actorId WHERE dob = MAX(dob)
 - C. INNER JOIN MovieActor b ON a.movieId = b.movieId INNER JOIN Actor c ON b.actorId = c.actorId WHERE dob in (SELECT MAX(dob) FROM Actor)
 - D. INNER JOIN MovieActor b ON a.movieId = b.movieId INNER JOIN Actor c ON b.actorId = c.actorId WHERE dob = (NOW()-dob)
34. What is the highest level of normalization in SQL?
- A. 2NF
 - B. 3NF
 - C. 4NF
 - D. 5NF
35. After reviewing the data, it was noticed that Jane was also in the movie Shape. What statement would add her to the Movie table.
- A. UPDATE MovieActor SET movieId=4, actorId=3
 - B. ADD movieId=4, actorId=3 TO MovieActor
 - C. INSERT INTO MovieActor VALUES (4,3)
 - D. CREATE NEW MovieActor VALUES (4,3)

36. What does the acronym DCL stand for?
- A. Database Consistency Limit
 - B. Data Creation Limitations
 - C. Database Cloud Link
 - D. Data Control Language
37. What is the difference between TRUNCATE and DELETE in SQL
- A. DELETE is a DCL command, TRUNCATE is a DML command
 - B. TRUNCATE data is recoverable, DELETE is not
 - C. TRUNCATE removes schema, table, constraints, and triggers; Delete removes just the data
 - D. DELETE can be used with a WHERE clause, TRUNCATE can not.
38. Referential Integrity refers to ____.
- A. The consistency of the Super Key in relation to the Primary Key
 - B. The accuracy and consistency of data within a relationship
 - C. Maintaining foreign keys between two tables
 - D. Having triggers on parent tables
39. A nonrelational database ____.
- A. Stores data in a non-tabular format
 - B. Cannot be queried
 - C. Is traditionally used for smaller data stores
 - D. Is best in non-cloud related applications.
40. Which of the following is not a main relationship type in relational database design?
- A. One-One
 - B. One-Many
 - C. Many-One
 - D. Many-Many
41. A data modeling technique that graphically illustrates information and relationships within a database is a(n) ____.
- A. ERD
 - B. IDE
 - C. DDM
 - D. SDD
42. Cardinality of a relationship in database terms refers to the number of ____ it contains.
- A. Indexes
 - B. Tuples
 - C. Keys
 - D. Columns

43. Which of the following is NOT an Index type in SQL?
- A. Hash
 - B. Clustered
 - C. Non-Clustered
 - D. Heap
44. Functions / methods that are saved in a database are called?
- A. Accessible Object
 - B. Database Functions
 - C. Dynamic Methods
 - D. Stored Procedures
45. The result of a CROSS JOIN is known as a ____.
- A. Relative Compliment of two tables
 - B. Cartesian Product
 - C. DDL Matrix
 - D. Differential Grid
46. Using a varchar(150) data field, how many characters would “Business Professional of America” typically take up?
- A. 32
 - B. 34
 - C. 75
 - D. 150
47. How would you empty the table “Business”, but leave the structure of the table intact?
- A. DROP TABLE Business
 - B. DELETE ALL FROM Business
 - C. TRUNCATE Business
 - D. EMPTY Business
48. How would you add a foreign key on other_Id from a table named other to a field main_Id in a related table named main?
- A. UPDATE other SET FOREIGN KEY ON other_Id = main.main_Id
 - B. MODIFY other ADD FOREIGN KEY ON other_Id = main.main_Id
 - C. ALTER TABLE other ADD FOREIGN KEY other ON other_Id = main.main_Id
 - D. ALTER TABLE other ADD CONSTRAINT otherToMain FOREIGN KEY (other_id) REFERENCES main (main_id)
49. The purpose of a rollback in SQL is to ____.
- A. To undelete data when using – DELETE ALL FROM table
 - B. To undo everything that was done in a Transaction
 - C. To Allow a DBA to undo database updates that have been committed
 - D. To maintain referential integrity

50. A disadvantage of OLAP would be its ____.
- A. Heavily dependent on IT
 - B. Non adaptive
 - C. Non dependable
 - D. Not Business Centered